Research Paper
Relationship Analysis

Generated 12 October 2026 at 10:16 UTC

CORPUS

papers

AT A GLANCE

Papers analysed	30
Entities extracted	1037
Bridge concepts	92
Knowledge graph	1159 nodes, 2047 edges
Document pairs scored	435
Contradictions	0 (0 confirmed)
Research gaps	5

HOW THIS WAS PRODUCED

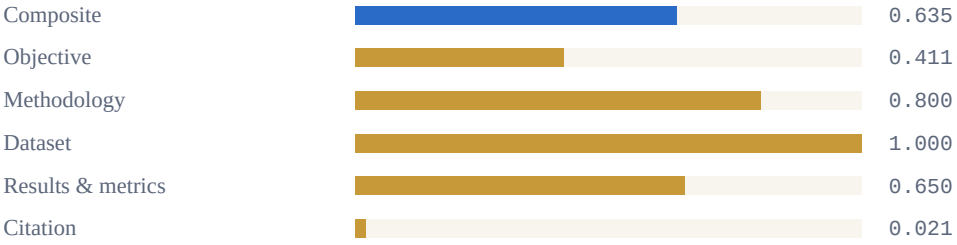
Extraction backend: heuristic. Embeddings: sentence-transformers. Completed in 69.54 seconds.

Every finding below carries an evidence trail: the verbatim sentence it was derived from, with its source document, section and page. Nothing in this report is generated prose about the papers - the quotations are the papers' own words.

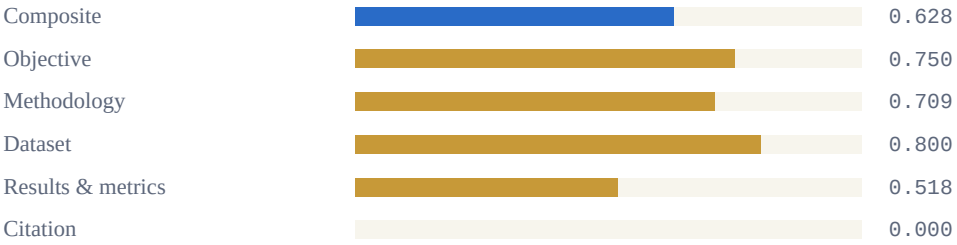
Strongest relationships

Each pair is scored across five dimensions - objective, methodology, dataset, results and citation overlap - and combined using the configured weights.

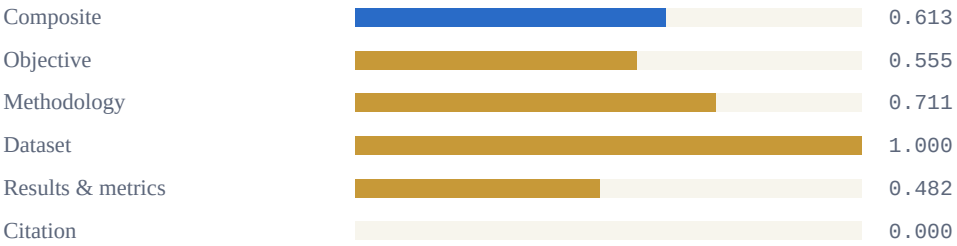
Masked Autoencoders Are Scalable Vision Learners · Swin Transformer: Hierarchical Vision Transformer using Shi·



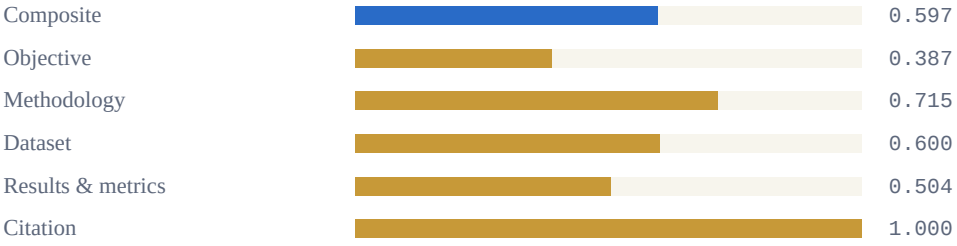
ALBERT: A LITE BERT FOR SELF-SUPERVISED LEARNING OF LANGUAG· · RoBERTa: A Robustly Optimized BERT Pretraining Approach



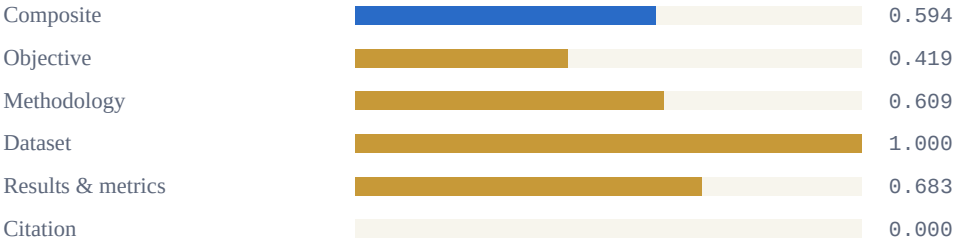
DistilBERT, a distilled version of BERT: smaller, faster, c· · ELECTRA: PRE-TRAINING TEXT ENCODERS AS DISCRIMINATORS RATHE·



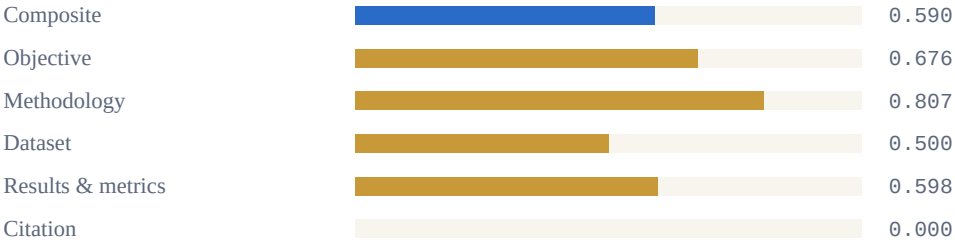
Momentum Contrast for Unsupervised Visual Representation Le· · Deep Residual Learning for Image Recognition



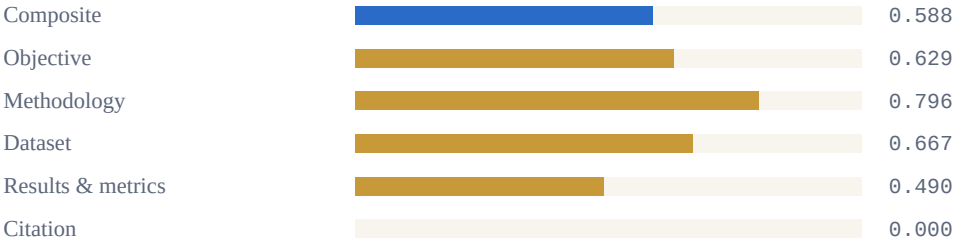
Bootstrap Your Own Latent A New Approach to Self-Supervised· · VERY DEEP CONVOLUTIONAL NETWORKS FOR LARGE-SCALE I MAGE REC·



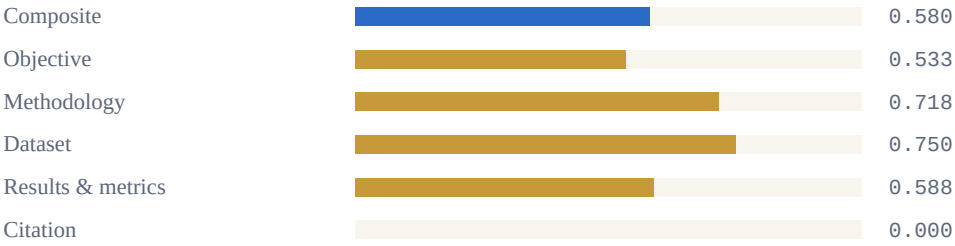
Squeeze-and-Excitation Networks · Wide Residual Networks



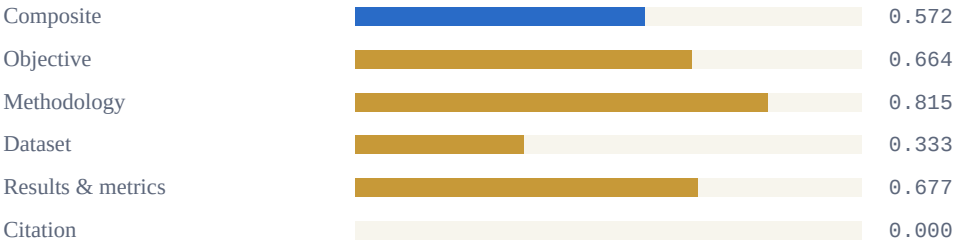
Deep Residual Learning for Image Recognition · Wide Residual Networks



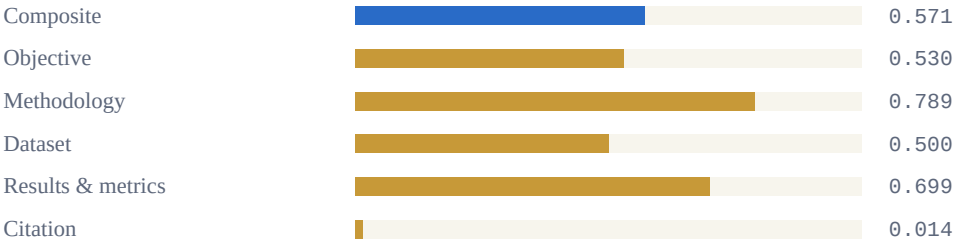
Deep Residual Learning for Image Recognition · Squeeze-and-Excitation Networks



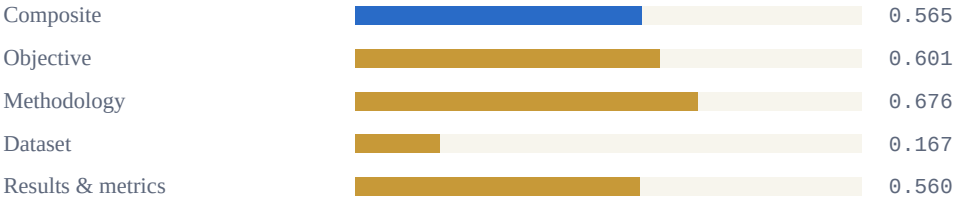
Momentum Contrast for Unsupervised Visual Representation Le· · A Simple Framework for Contrastive Learning of Visual Repre·



Dense Passage Retrieval for Open-Domain Question Answering · Retrieval-Augmented Generation for Knowledge-Intensive NLP·

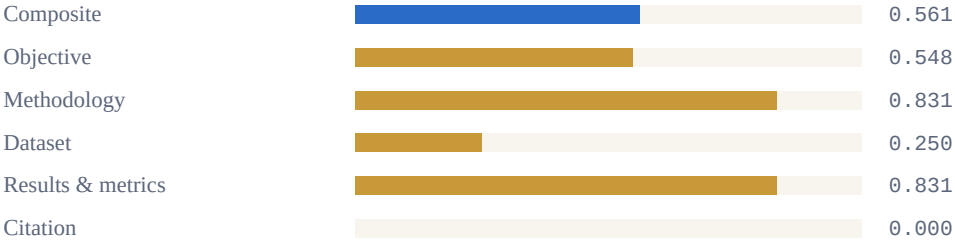


VERY DEEP CONVOLUTIONAL NETWORKS FOR LARGE-SCALE I MAGE REC· · Wide Residual Networks





Bootstrap Your Own Latent A New Approach to Self-Supervised· · A Simple Framework for Contrastive Learning of Visual Repre·



Contradictions

A pair is reported only when both papers state a measured result for the same metric on the same dataset, under the same evaluation regime, and the values differ materially. Papers studying different methods rarely meet that bar, so an empty section is a real result rather than a failure.

No contradictions were found in this corpus.

Research gaps

Ranked by importance: how well evidenced the gap is, how specific it is, and whether a paper states it outright rather than the analysis inferring it. Only the strongest are listed.

GAP 1 · IMPORTANCE 0.91

Empirical results show that DocRED is challenging for existing RE methods, which indicates that document-level RE remains an open problem and requires further efforts.

Supporting papers: DocRED: A Large-Scale Document-Level Relation Extract·

> *·Empirical results show that DocRED is challenging for existing RE methods, which indicates that document-level RE remains an open problem and requires further efforts.·*
> `[docred_2019 · §abstract · p.1]`

GAP 2 · IMPORTANCE 0.91

Experimental results show that the performance of existing methods declines significantly on DocRED, indicating the task documentlevel RE is more challenging than sentence-level RE and remains an open problem.

Supporting papers: DocRED: A Large-Scale Document-Level Relation Extract·

> *·Experimental results show that the performance of existing methods declines significantly on DocRED, indicating the task documentlevel RE is more challenging than sentence-level RE and remains an open problem.·*
> `[docred_2019 · §introduction · p.1]`

GAP 3 · IMPORTANCE 0.91

Additionally, providing provenance for their decisions and updating their world knowledge remain open research problems.

Supporting papers: Retrieval-Augmented Generation for Knowledge-Intensiv·

> *·Additionally, providing provenance for their decisions and updating their world knowledge remain open research problems.·*
> `[rag_2020 · §abstract · p.1]`

GAP 4 · IMPORTANCE 0.73

This however is computationally expensive, and the high level of detail required for image generation may not be necessary for representation learning.

Supporting papers: Bootstrap Your Own Latent A New Approach to Self-Supe·

> *·This however is computationally expensive, and the high level of detail required for image generation may not be necessary for representation learning.·*
> `[byol_2020 · §related_work · p.2]`

GAP 5 · IMPORTANCE 0.73

Context-less example-based learning has the drawback that many different hypotheses can be consistent with the data, especially in the one-shot case.

Supporting papers: Learning Transferable Visual Models From Natural Lang·

> *·Context-less example-based learning has the drawback that many different hypotheses can be consistent with the data, especially in the one-shot case.·*
> `[clip_2021 · §results · p.8]`

Corpus

30 papers, with the category assigned by the classifier and the sections recovered during segmentation.

ALBERT: A LITE BERT FOR SELF-SUPERVISED LEARNING OF LANGUAGE REPRESENTATIONS
`Experimental (confidence 0.82) · 8 sections · 17 pages · 2 references`
abstract, introduction, related_work, methodology, experiments, results, conclusion, references

Attention Is All You Need
`Experimental (confidence 0.71) · 7 sections · 15 pages · 42 references`
abstract, introduction, related_work, methodology, results, conclusion, references

Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate·
`Experimental (confidence 0.56) · 6 sections · 11 pages · 4 references`
abstract, introduction, methodology, experiments, conclusion, references

BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding
`Experimental (confidence 0.70) · 8 sections · 16 pages · 4 references`
abstract, introduction, related_work, methodology, experiments, results, conclusion, references

Bootstrap Your Own Latent A New Approach to Self-Supervised Learning
`Experimental (confidence 0.62) · 8 sections · 35 pages · 98 references`
abstract, introduction, related_work, methodology, experiments, results, conclusion, references

Learning Transferable Visual Models From Natural Language Supervision
`Experimental (confidence 0.70) · 8 sections · 48 pages · 4 references`
abstract, introduction, methodology, experiments, results, conclusion, related_work, references

Training data-efficient image transformers & distillation through attention
`Experimental (confidence 0.69) · 6 sections · 22 pages · 61 references`
abstract, introduction, related_work, experiments, conclusion, references

Densely Connected Convolutional Networks
`Experimental (confidence 0.55) · 6 sections · 9 pages · 37 references`
abstract, introduction, related_work, experiments, conclusion, references

DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter
`Experimental (confidence 0.50) · 7 sections · 5 pages · 1 references`
abstract, introduction, results, experiments, related_work, conclusion, references

DocRED: A Large-Scale Document-Level Relation Extraction Dataset
`Experimental (confidence 0.64) · 8 sections · 14 pages · 1 references`
abstract, introduction, experiments, methodology, results, related_work, references, conclusion

Dense Passage Retrieval for Open-Domain Question Answering
`Experimental (confidence 0.73) · 8 sections · 13 pages · 4 references`
abstract, introduction, related_work, experiments, results, methodology, conclusion, references

EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks
`Experimental (confidence 0.75) · 7 sections · 11 pages · 29 references`
abstract, introduction, related_work, methodology, experiments, conclusion, references

ELECTRA: PRE-TRAINING TEXT ENCODERS AS DISCRIMINATORS RATHER THAN GENERATORS
`Experimental (confidence 0.83) · 7 sections · 18 pages · 5 references`
abstract, introduction, methodology, experiments, related_work, conclusion, references

A Survey on Knowledge Graphs: Representation, Acquisition and Applications
`Experimental (confidence 0.38) · 5 sections · 27 pages · 36 references`
abstract, experiments, introduction, conclusion, references

Masked Autoencoders Are Scalable Vision Learners
`Experimental (confidence 0.82) · 8 sections · 14 pages · 61 references`
abstract, introduction, methodology, related_work, conclusion, references, experiments, results

MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications
Experimental (confidence 0.62) · 6 sections · 9 pages · 26 references
abstract, related_work, introduction, methodology, conclusion, references

Momentum Contrast for Unsupervised Visual Representation Learning
Experimental (confidence 0.58) · 8 sections · 12 pages · 63 references
abstract, introduction, methodology, related_work, experiments, results, conclusion, references

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks
Experimental (confidence 0.75) · 8 sections · 19 pages · 67 references
abstract, introduction, methodology, experiments, results, related_work, conclusion, references

Deep Residual Learning for Image Recognition
Experimental (confidence 0.58) · 6 sections · 12 pages · 48 references
abstract, introduction, related_work, experiments, results, references

RoBERTa: A Robustly Optimized BERT Pretraining Approach
Experimental (confidence 0.57) · 6 sections · 13 pages · 1 references
abstract, introduction, related_work, experiments, conclusion, references

SCI BERT: A Pretrained Language Model for Scientific Text
Experimental (confidence 0.60) · 8 sections · 6 pages · 1 references
abstract, introduction, methodology, experiments, results, conclusion, references, related_work

A Simple Framework for Contrastive Learning of Visual Representations
Experimental (confidence 0.58) · 8 sections · 20 pages · 36 references
abstract, introduction, methodology, results, experiments, related_work, conclusion, references

SPECTER: Document-level Representation Learning using Citation-informed Transformers
Experimental (confidence 0.69) · 8 sections · 13 pages · 1 references
abstract, introduction, related_work, methodology, experiments, results, references, conclusion

Squeeze-and-Excitation Networks
Experimental (confidence 0.67) · 7 sections · 13 pages · 43 references
introduction, related_work, conclusion, methodology, results, experiments, references

Swin Transformer: Hierarchical Vision Transformer using Shifted Windows
Experimental (confidence 0.64) · 8 sections · 14 pages · 83 references
abstract, introduction, related_work, methodology, experiments, results, conclusion, references

Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer
Experimental (confidence 0.68) · 7 sections · 67 pages · 1 references
abstract, introduction, experiments, methodology, results, conclusion, references

VERY DEEP CONVOLUTIONAL NETWORKS FOR LARGE-SCALE I MAGE RECOGNITION
Experimental (confidence 0.80) · 7 sections · 14 pages · 4 references
abstract, introduction, methodology, conclusion, experiments, results, references

AN I MAGE IS WORTH 16 X 16 WORDS: TRANSFORMERS FOR I MAGE RECOGNITION AT SCALE
Experimental (confidence 0.75) · 8 sections · 22 pages · 4 references
abstract, introduction, related_work, methodology, experiments, results, conclusion, references

Wide Residual Networks
Experimental (confidence 0.58) · 5 sections · 15 pages · 28 references
abstract, introduction, experiments, conclusion, references

XLNet: Generalized Autoregressive Pretraining for Language Understanding
Experimental (confidence 0.60) · 8 sections · 18 pages · 39 references
abstract, introduction, methodology, related_work, conclusion, experiments, results, references